CapUBS: Capsule Network-Based Band Selection for Underwater Hyperspectral Imagery

Qi Li, Junwen Wang, Xingyuan Zu, Dan Chen, Ke Zhang, and Jinghua Li

***Abstract*— Underwater hyperspectral imagery (HSI) finds extensive applications in critical domains such as marine target detection. However, spectral offset caused by underwater light propagation presents significant challenges for existing hyperspectral band selection methods, which often fail to capture interband correlations in underwater environments effectively. To address this issue, this article proposes a novel underwater hyperspectral band selection method based on a capsule network (CapUBS) designed to mitigate spectral offset variations and reduce redundancy in band selection for underwater HSIs. Especially, the dual-stream latent variable encoding (DLVE) module takes the original and averaged spectra as inputs and employs deep feature extraction blocks (DFEBs) to capture local details and global semantic features, thereby alleviating spectral variability. The capsule network-based band characterization (CNBC) module utilizes a dynamic routing mechanism to generate digital capsules representing joint multiband features. This module characterizes the multidimensional spectral attributes via vector directions and quantifies the relative importance of spectral bands using vector norms. The feature separation factor evaluation (FSFE) module computes the initial band importance ranking to refine the selection process further. It evaluates a feature separation factor to identify key bands with minimal redundancy and high representativeness, resulting in an optimal subset of spectral bands. Finally, the key band feature reconstruction (KBFR) module leverages these selected bands to reconstruct the full-band spectrum accurately. Comprehensive quantitative and qualitative analyses conducted on four classical hyperspectral datasets and underwater hyperspectral experimental datasets demonstrate the superior performance of the proposed method compared to state-of-the-art techniques. The code will be available at https://github.com/lq132069/CapUBS**

***Index Terms*— Band selection, capsule network, deep learning, hyperspectral imagery (HSI).**

Received 12 January 2025; revised 4 March 2025 and 13 April 2025; accepted 29 April 2025. Date of publication 2 May 2025; date of current version 15 May 2025. This work was supported by the Science and Technology on Electromechanical Dynamic Control Laboratory under Grant 6142601240202. *(Corresponding author: Jinghua Li.)*

Qi Li, Junwen Wang, and Xingyuan Zu are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: lq132069@163.com; wjwwjw2002@hotmail.com; zuxingyuan@outlook.com).

Dan Chen is with the School of Automation and Information Engineering, Xi'an University of Technology, Xi'an 710048, China (e-mail: chdh@xaut.edu.cn).

Ke Zhang is with the Science and Technology on Electromechanical Dynamic Control Laboratory, Xi'an 710065, China (e-mail: 32578792@qq.com).

Jinghua Li is with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710072, China, and also with the Science and Technology on Electromechanical Dynamic Control Laboratory, Xi'an, China (e-mail: ljhy6331@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3566399

I. Introduction

HYPERSPECTRAL imaging technology captures detailed spectral information across continuous bands in the visible to near-infrared range, serving as a critical tool for target detection [1], [2], [3], environmental monitoring [4], resource assessment [5], and super-resolution [6]. Hyperspectral imaging can detect subtle spectral differences compared to traditional optical sensors, making it particularly suitable for identifying and classifying underwater targets. The complex underwater environment presents significant challenges for applying hyperspectral data [7]. Light propagation in water is influenced by scattering and absorption, causing wavelength-dependent spectral attenuation [8]. Additionally, chlorophyll-rich waters shift the light spectrum, distorting spectral features [9]. Hyperspectral data typically consists of hundreds of bands, each rich in information but often containing redundancy. This redundancy increases storage and computational burdens and may hinder feature extraction [10]. Reducing redundancy while preserving key information is a critical challenge in underwater hyperspectral data processing. Band selection is a key approach to address this issue by selecting an optimal set of bands, reducing dimensionality and redundancy, and enhancing the efficiency and accuracy of subsequent analyses. Therefore, it is necessary to develop the band selection method for underwater hyperspectral imageries (HSIs).

Traditional hyperspectral band selection methods rely on statistical analysis and classical feature extraction techniques. These methods reduce computational complexity by performing dimensionality reduction and keyband screening while preserving essential information in HSIs [11]. Among classical unsupervised methods, algorithms that combine spatial and spectral features offer significant advantages. Wang et al. [12] proposed the fast hyperspectral band selection (FNGBS) method, which achieves high efficiency by mining contextual information over an extensive spectral range using a coarse–fine strategy. Tang et al. [13] employed a self-representation model that preserves spatial and spectral structure, enabling structured representation of hyperspectral data via superimage segmentation and spectral correlation learning. This approach offers a novel perspective for unsupervised band selection. Additionally, band selection methods based on geometric features further expand the technical approaches of traditional methods. Zhang et al. [14] proposed the MEV-SFS method, combining maximum ellipsoidal

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

volume (MEV) and sequential forward search (SFS). They developed the orthogonal projection-based band selection (OPBS) method, which effectively utilizes spectral geometric features to select key bands. Jia et al. [16] improved the fast density peak-based clustering (FDPC) algorithm [15] and proposed a novel hyperspectral band selection method based on ranked clustering, optimizing its applicability for hyperspectral band selection. The band prioritization-based selection method prioritizes spectral bands based on BP criteria, including variance [17], mutual information [18], [19], information entropy (IE) [20], information dispersion [21], maximum variance principal component analysis (MVPCA) [22], constrained band selection (CBS) [23], and minimum estimated abundance covariance (MEAC) [24]. Wang et al. [25] proposed an optimal clustering framework (OCF) for identifying the best clustering structure in HSIs. The classical methods IE, MVPCA, and FDPC are optimized, and the NC-OC-IE, NC-OC-MVPCA, and TRC-OC-FDPC algorithms are introduced. However, these methods have limitations when dealing with nonlinear spectral relationships and noise interference in hyperspectral data, particularly in complex underwater environments.

Unsupervised hyperspectral band selection methods based on deep learning have recently gained significant attention due to the rapid development of deep learning technology [26], [27], [28]. These methods extract deep features from spectral data using deep network models [29], [30], [31], [32] and combine them with feature importance assessment for band selection [33], demonstrating superior performance and flexibility. Different deep learning methods offer unique characteristics and provide various solutions for the hyperspectral band selection problem. Feng et al. [34] proposed the multi-intelligent body DRL (MH-DRL) method, which incorporates hybrid teacher guidance to enhance the efficiency of band selection in multi-intelligent scenarios. To address the challenge of deep fusion of spatial and spectral information in hyperspectral data [35], Zhang et al. [36] designed the spatial–spectral full-attention network (SSFAN), which captures global spatial–spectral information efficiently through the spatial–spectral sample generator (SSSG) and nearest-neighbor scoring (NNS) modules, improving model performance. Studies based on convolutional neural networks (CNNs) have expanded the application of deep learning in band selection [37], [38], [39]. Ribalta Lorenzo et al. [39] proposed an attention-based CNN (ABCNN) to extract key spectral features by weighting band importance in anomaly detection tasks. Cai et al. [37] took hyperspectral band selection as a spectral reconstruction task. They proposed the unified BS-Net framework, which leverages the attention mechanism to optimize the band selection process. Dou et al. [40] proposed an attention-based autoencoder model to model potential nonlinear dependencies between bands, assuming HSIs can be sparsely reconstructed. Additionally, Roy et al. [41] developed the end-to-end dual-attention reconstruction network (DARecNet), which extracts nonlinear contextual information via positional and channel attention modules and a multidimensional CNN, seamlessly combining band selection and feature extraction. While deep learning methods have shown strong capabilities in hyperspectral band selection, particularly in spectral feature extraction and model optimization, they must improve robustness in complex environments.

Few existing band selection methods have been designed for underwater hyperspectral imaging. Qi et al. [42] proposed a bathymetric-based band selection method, which integrates an underwater optical attenuation model into the band selection process. By learning a latent spectral representation, their approach constructs a physically meaningful subspace where targets are optimally distinguishable from the background. Fu et al. [43] introduced a band selection method based on the constrained target optimal index factor, which selects a subset of spectral bands that exhibit strong responses to the target, thereby enhancing detection performance. However, these methods do not explicitly account for the impact of water scattering characteristics on band selection.

Band selection for underwater HSI faces several challenges. First, underwater spectral distortion significantly complicates the band selection process. The light attenuation coefficient in water varies with wavelength, causing the same object to exhibit different spectral characteristics at different depths or environments. Additionally, chlorophyll in the water column induces spectral shifts in light environments [44], distorting spectral features. Traditional methods struggle to adapt to these dynamic changes. Second, water absorbs and scatters light differently across wavelengths [45], [46], leading to rapid energy attenuation in most bands and slower attenuation in the bluish-green bands [47]. With their high transmittance and signal-to-noise ratio, these bands are the primary source of underwater spectral information, making the bluish-green region the focus of band selection. Finally, hyperspectral data contain a significant proportion of redundant bands. While hyperspectral data contain rich information, not all bands contribute equally to a specific task; some are noisy or irrelevant. These bands waste storage and computational resources and may negatively impact model performance. Studying band selection methods for underwater HSIs is essential to address spectral shifts, focus on the bluish-green bands, reduce redundancy, and enhance feature extraction effectiveness.

To address spectral offset variation and redundancy, this article proposes a capsule network-based underwater HSI band selection method in an unsupervised learning framework named CapUBS. The CapUBS first extracts local details and global semantic features using a dual-stream latent variable encoding (DLVE) module, with the original and average spectra as inputs. This suppresses feature learning path bias caused by underwater spectral variations. Next, the capsule network-based band characterization (CNBC) module maps latent variables into primary capsules, generates digital capsules through dynamic routing, and characterizes spectral attributes using vector directions. The norm of the vector is applied to assess band importance, enabling fine-grained extraction and quantification of joint multiband features and effectively reducing spectral offset. The feature separation factor evaluation (FSFE) module calculates feature separation factors, ensuring that selected bands reduce redundancy while retaining representativeness, resulting in an optimized set

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

of key bands. Finally, the key band feature reconstruction (KBFR) module recovers full-band information through feature inversion and upsampling based on key band features. In summary, the main contributions of this article can be listed as the following.

1) We innovatively apply capsule networks to underwater hyperspectral band selection, integrating a DLVE module. Local details and global semantic features are gradually extracted through multiple deep feature extraction blocks (DFEBs), and high-dimensional latent variable distributions are generated. These distributions are guided by average spectral features, effectively suppressing spectral variations caused by underwater environmental disturbances.
2) The digital capsule in the CNBC module represents joint multiband features, with its vector direction capturing spectral multidimensional attributes and inhibiting underwater spectral offset. The norm quantifies the initial band importance. The FSFE module ranks the importance of the spectral bands and filters key bands with low redundancy and high representativeness to obtain an optimal band subset.
3) Comprehensive comparison experiments were conducted on four classical and underwater hyperspectral datasets. The results demonstrate that our proposed method outperforms existing mainstream band selection methods, highlighting its significant advantages and broad applicability.

The remainder of this article is structured as follows. Section II delves into related work detailing existing research in the field. Section III provides a detailed explanation of the model proposed in this article. Section IV presents the experimental results and related analysis with the hyperparameter analysis. Finally, Section V concludes this article.

II. Related Work

In this section, we will provide an overview of the capsule network, focusing on the HSI processing algorithm based on it. Additionally, we will outline the motivation for the band selection method for underwater HSIs based on the capsule network.

A. Capsule Network and HSIs

Capsule network [48], a novel deep learning architecture, has recently made significant advances in computer vision, particularly in feature representation and spatial relationship capture. Traditional CNNs, while effective in feature extraction, fail to capture complex spatial structures and relationships between features due to limitations in localization and pooling operations. Capsule networks encode features through vectorization, representing the existence of features and characterizing their attributes (e.g., location and orientation), providing a natural advantage in capturing spatial information and pattern relationships. As capsule networks have developed, many researchers have applied them to HSI processing to overcome the shortcomings of traditional CNN methods in hyperspectral data processing.

Capsule networks have demonstrated powerful feature extraction and spatial pattern mining capabilities in HSI processing. Zhu et al. [49] proposed the Conv-Capsule network, which enhances the capsule network by reducing trainable parameters and significantly improves performance in HSI classification. Wang et al. [50] combined the capsule network with the fully connected neural network, improving HSI classification performance, particularly under conditions of insufficient labeled samples. Additionally, Zhai and Zhao [51] proposed the two-stream spectral–spatial convolutional capsule network (TSCCN), which achieved a breakthrough in small-sample classification by utilizing the rich spectral–spatial information in HSIs. Sun et al. [52] proposed an HSI segmentation model based on extended multimorphic attribute profile (EMAP) features and a cubic capsule network, overcoming the limitations of traditional CNNs in capturing rotating targets and fine spatial features. Deng et al. [53] proposed an improved two-layer capsule network for the HSI classification with limited training samples, maintaining high classification accuracy despite sample scarcity. Several approaches have improved the capsule network-based architecture [54], [55], [56], [57], including the unsupervised multibranch capsule network by Xu et al. [58], the capsule attention network for HSI classification by Wang et al. [59], and the hierarchical capsule network by Shi et al. [60], all of which demonstrate strong capabilities in HSI processing.

Although capsule networks have demonstrated excellent performance in classifying, segmenting, and feature extraction of HSIs, their application to the band selection problem remains underdeveloped. The utilization of capsule networks to address issues such as spectral shift, redundancy, and data imbalance in hyperspectral band selection remains largely unexplored. Therefore, capsule network-based band selection methods, particularly in the context of complex underwater HSIs, hold significant research value and application potential.

B. Motivation

With the complexity of underwater environments, band selection for underwater HSIs faces numerous challenges. As shown in Fig. 1, there are significant differences between land-based HSIs and underwater HSIs. The complex aquatic environment, variations in underwater spectral offsets, and issues of redundancy greatly diminish the effectiveness of traditional band selection methods for HSIs. The spectral properties of aquatic environments are influenced by water environment, suspended particulate matter, and chlorophyll, resulting in spectral offset shifts and information distortion. Traditional spectral angle-based selection methods struggle to adapt to these dynamic changes. Consequently, ensuring the representativeness of spectral information while eliminating redundant bands and effectively addressing underwater spectral offset variation has become a critical challenge in HSI processing.

Motivated by the strong feature representation capabilities of capsule networks, this article proposes a capsule network-based method for band selection in underwater HSIs.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 1. Pseudocolor images. (a) Land-based HSI. (b) Underwater HSI.

Digital capsules in capsule networks capture spatial relationships and multidimensional attributes in high-dimensional spectral data through vectorized feature representations, effectively addressing the challenges of spectral offset variance in underwater environments. The proposed method overcomes the limitations of traditional approaches, which fail to handle the complexity of underwater spectra, by employing a DLVE module that integrates local details with global semantic feature learning. Additionally, leveraging digital capsules' joint multiband vector characterization capability, this method quantifies the importance of underwater spectral bands more accurately. It evaluates band redundancy through feature separation factors, thus optimizing the band selection process and enhancing efficiency and accuracy.

Unlike existing methods, the approach presented in this article integrates the advantages of vector feature extraction with capsule networks to develop a band selection method for underwater HSIs. Notably, this is the first application of capsule networks to underwater hyperspectral band selection. This work not only broadens the application of capsule networks in hyperspectral band selection but also offers new insights into the efficient utilization of underwater HSIs.

III. Methodology

In this section, we will explain the capsule network-based band selection method for underwater HSIs, named CapUBS, as illustrated in Fig. 2. The CapUBS employs a self-encoder architecture, categorized under unsupervised learning, and primarily consists of the following modules: DLVE module, CNBC module, FSFE module, and KBFR module. First, the DLVE module takes the original and average spectra as inputs, extracting local and global semantic features through multiple DFEBs to generate high-dimensional latent variable distributions. The average spectrum features guide the learning of the original spectral features, effectively mitigating spectral variations caused by environmental interference. Next, the CNBC module maps the latent variables into primary capsules and aggregates them via the dynamic routing strategy into digital capsules representing multiple bands' joint features. The vector direction of the digital capsules describes the spectral multidimensional attributes, effectively suppressing underwater spectral offset. Their norm is used to quantify the initial band importance, followed by a nonlinear reconstruction of the spectral latent variables based on the digital capsules. The FSFE module then filters key bands with low redundancy and high representativeness by calculating the underwater spectral bands' initial importance ranking and feature separation factors, thereby obtaining the optimal subset of bands. Finally, using the key band feature, the KBFR module effectively reconstructs the original full-band spectra.

A. DLVE Module

The input to the CapUBS is the raw underwater hyperspectral data cube $X \in \mathbb{R}^{C\times P\times P}$, where C and P represent the number of spectral channels and the width of the data cube, respectively. The average spectrum $X_a \in \mathbb{R}^{C\times P\times P}$ is obtained by applying an average filter with a specified kernel size 3×3. The DLVE module consists of two branches, each taking the original and average spectra as inputs. It gradually extracts multilevel feature representations of the underwater HSI through multiple DFEBs. This process focuses on the local spectra details and retains global semantic information, generating a high-dimensional latent variable distribution representation. Notably, the deep features of the averaged spectra guide the learning process of the original spectral features, effectively suppressing spectral variations caused by the complex underwater environment.

1) Deep Feature Extraction Block: The DFEB block is designed to efficiently capture local details, global contextual relationships, and multiscale feature representations in the input underwater HSIs, enhancing the feature representation capability for downstream tasks. The module is implemented in three phases: local feature extraction, attentional feature computation, and global feature enhancement, as illustrated in Fig. 3.

Local features from the input feature F_i are first extracted using a 3×3 convolutional layer to generate a preliminary spatial representation, emphasizing feature patterns in the local neighborhood

$$X_{\text{conv1}} = \text{ReLU}(W_{\text{conv1}} * F_i + b_{\text{conv1}}) \tag{1}$$

where $*$ denotes the convolution operation and W_{conv1} and b_{conv1} are the convolution kernel weights and bias terms, respectively.

To further capture the long-range dependencies in the input feature, the module employs the attention mechanism to generate the query matrix Q, key matrix K, and value matrix V using three parallel 3×3 convolutions. Attention weights are computed by performing a dot-product operation between Q and K, which measures the relationship between different spatial locations. Subsequently, the attention weight matrix is multiplied by the value matrix V to generate the weighted features F_{attn}

$$F_{\text{attn}} = \text{softmax}\left(\frac{Q_v^\top K_v}{\sqrt{C}}\right)V \tag{2}$$

where Q_v and K_v are vector forms extracted from Q and K through the spreading operation, respectively.

Based on the generated attention features, the module further compresses and processes the spectral features using a 1×1 convolution operation. It then concatenates them with the original local features to enhance the spectral feature representation. The concatenated spectral features are projected to the input feature dimensions through another 1×1 convolution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

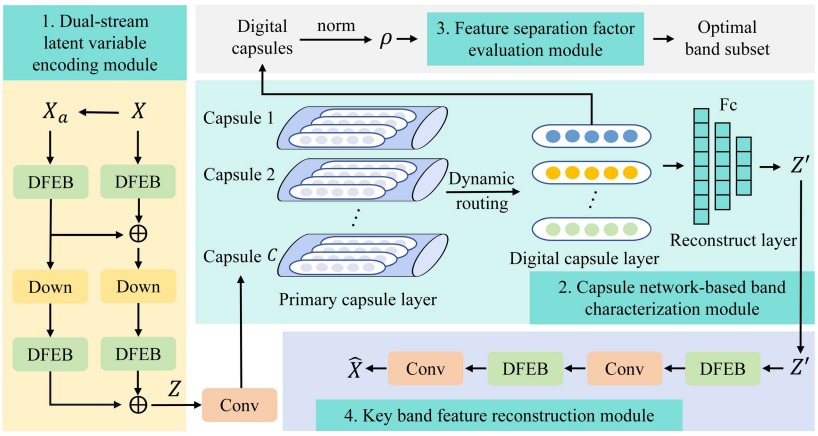


Fig. 2. Framework of the CapUBS. First, the DLVE module takes the original and average spectra as inputs, extracting local and global semantic features through multiple DFEB blocks to generate high-dimensional latent variable distributions Z. Next, the CNBC module then maps these latent variables Z into primary capsules. Through dynamic routing, these primary capsules are aggregated into digital capsules representing joint features of multiple bands, forming the key band feature Z'. Then, using these digital capsules, the FSFE module filters key bands with low redundancy and high representativeness. Finally, the KBFR module reconstructs the original full-band spectra based on the key band feature Z'.

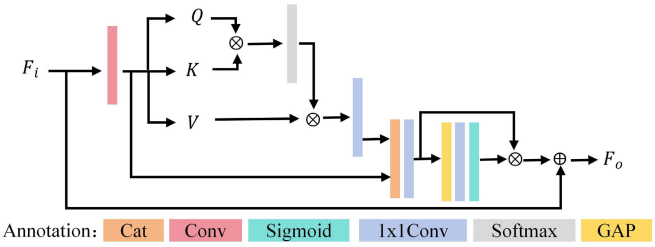


Fig. 3. Structure of the DFEB module.

layer

$$F_{\text{cat}} = \text{Cat}(X_{\text{conv1}}, \text{ReLU}(W_{\text{conv2}} * F_{\text{attn}}))$$
$$F_f = W_{\text{conv3}} * F_{\text{cat}} \tag{3}$$

where $\text{Cat}(\cdot)$ denotes the concatenation operation.

The DFEB introduces a global spectral feature enhancement mechanism to enhance its ability to capture global spectral features. First, the fused spectral features F_f are downsampled using global average pooling to generate a global descriptor vector F_{gap}. Then, a scaling factor is generated using a 1×1 convolutional layer followed by a Sigmoid activation function $\sigma(\cdot)$:

$$S = \sigma\left(W_{\text{gap}} * F_{\text{gap}}\right) S = \sigma\left(W_{\text{gap}} * F_{\text{gap}}\right). \tag{4}$$

The generated scaling factor S is resized to match the dimensions of the input features and then element-wise multiplied with the fused features to adjust their importance. The final adjusted features are combined with the input features through a residual connection

$$F_o = F_i + F_f \cdot S. \tag{5}$$

The DFEB block enhances the underwater spectral feature representation by effectively suppressing redundant information, capturing feature interactions through attention and global enhancement mechanisms, and preserving global and local information from the input features.

2) DLVE Architecture: The DLVE module extracts latent variable features using a dual-stream architecture based on the DFEB block. It separately models the original and average spectra and facilitates information interaction and enhancement through the feature fusion mechanism. The average spectral branch guides the entire spectral feature extraction process, with its depth features effectively regulating the learning of the original spectral branch to mitigate interference from the complex underwater environment.

The two branches receive the original X and averaged spectra X_a as inputs and encode multilevel spectral features using the DFEB block. The DFEB block first processes each branch's input to obtain intermediate feature representations $\text{DFEB}(X)$ and $f_1(X_a)$

$$f_1(X_a) = \text{DFEB}(X_a) \tag{6}$$
$$f_1(X) = \text{DFEB}(X) + f_1(X_a). \tag{7}$$

Next, the features of the original spectral branch are weighted and fused with those of the averaged spectral branch to obtain $f_1(X)$, followed by downsampling of $f_1(X)$ and $f_1(X_a)$. Specifically, the input features are first processed by a convolutional layer, followed by a LeakyReLU activation function and another convolutional layer. Simultaneously, the input features are processed by a parallel convolutional layer. Finally, the results of these two branches are summed along the channel dimension to obtain the downsampled features $f_2(X)$ and $f_2(X_a)$

$$f_2(X_a) = \text{Down}(f_1(X_a) \tag{8}$$
$$f_2(X) = \text{Down}(f_1(X)). \tag{9}$$

The features from both branches are further extracted from the deeper layers of the original and averaged spectral branches through the DFEB block and fused to obtain the latent variable representation Z

$$Z = \text{DEFB}(f_2(X_a)) + \text{DEFB}(f_2(X)). \tag{10}$$

The DLVE module integrates the complementary advantages of two branches, allowing the output latent variables

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

to capture both the original spectrum's detailed characteristics and the average spectrum's global semantics. The deep features of the average spectrum guide the learning process of the original spectral features, effectively mitigating spectral variations caused by the complex underwater environment.

B. CNBC Module

Due to the effects of absorption, scattering, refraction, and spectral shifts in the underwater environment, the information of a single band becomes impure. It may be represented by the combined nonlinearities of multiple adjacent bands. To address this, the capsule vector form is used to characterize the essential features of the spectral bands. The CNBC module aims to extract joint multiband features and their potential correlations from underwater HSIs, leveraging the structural advantages of capsule networks. The CNBC module uses the latent variables Z of the input spectra, projecting them into C primary capsules $\{\mathbf{u_i}\}_{i=1}^{C}$, where C is the number of bands. Each primary capsule $\mathbf{u_i}$ represents a latent variable of one band. The formula for constructing the primary capsules is as follows:

$$\mathbf{u_i} = f(Z) \tag{11}$$

where $f(\cdot)$ denotes the convolution function that controls the nonlinear projection of the latent variable into each primary capsule.

The primary capsules $\{\mathbf{u_i}\}_{i=1}^{C}$ are aggregated into digital capsules $\{\mathbf{v_j}\}_{j=1}^{C}$ through the dynamic routing strategy. Each digital capsule $\mathbf{v_j}$ captures the deep semantic information of multiband joint features and band correlations. The dynamic routing mechanism first calculates the coupling coefficient c_{ij} from each primary capsule $\mathbf{u_i}$ to the corresponding digital capsules $\mathbf{v_j}$

$$c_{ij} = \frac{\exp(b_{ij})}{\sum_{k=1}^{C} \exp(b_{ik})} \tag{12}$$

where b_{ij} denotes the coupling weight, the initial value is set to 0, and is gradually optimized in the dynamic routing process by the following formula:

$$b_{ij} = b_{ij} + \mathbf{u_i} \cdot \mathbf{v_j}. \tag{13}$$

The digital capsule $\mathbf{v_j}$ then aggregates information from the primary capsule $\mathbf{u_i}$ through a weighted sum

$$\mathbf{s_j} = \sum_{i=1}^{n} c_{ij} \mathbf{u_i} \tag{14}$$

$$\mathbf{v_j} = \text{squash}(\mathbf{s_j}) = \frac{\|\mathbf{s_j}\|^2}{1 + \|\mathbf{s_j}\|^2} \frac{\mathbf{s_j}}{\|\mathbf{s_j}\|} \tag{15}$$

where squash$(\cdot)$ is a nonlinear compression function specific to the capsule network, used to normalize the norm of the digital capsule vector, ensuring that its value lies between [0, 1].

After three dynamic routing processes, the digital capsules $\{\mathbf{v_j}\}_{j=1}^{C}$ are obtained. The vector direction of the digital capsule $\mathbf{v_j}$ represents the multidimensional feature attributes of the spectral band, while its norm quantifies the initial band importance ρ_j. Subsequently, leveraging the nonlinear characterization capability of the fully connected layer, the digital capsule further reconstructs the spectral latent variable Z', integrating the joint features across multiple bands and their respective importance for the subsequent recovery of the original input spectrum

$$\rho_j = \|\mathbf{v_j}\|. \tag{16}$$

The dynamic routing mechanism of the capsule network accurately characterizes the initial band importance $\{\rho_j\}_{j=1}^{C}$. Each digital capsule captures the joint features across multiple bands and the deep semantic information of their correlations.

C. FSFE Module

To reduce spectral redundancy, the FSFE module ranks the initial band's importance $\{\rho_j\}_{j=1}^{C}$ and sequentially calculates the feature separation factor for each band. A comprehensive evaluation factor is then generated by a weighted fusion of the initial underwater spectral band's importance and the feature separation factor. This factor is then used to filter key bands with low redundancy.

The FSFE module first sorts the bands in descending order based on their initial importance ρ_j, as reflected by the digital capsule norm, forming a sorted list $\{\rho_{(1)}, \rho_{(2)}, \ldots, \rho_{(C)}\}$, where $\rho_{(i)} \geq \rho_{(i+1)}$. After sorting, the set of band indices is denoted as $\{\pi(1), \pi(2), \ldots, \pi(C)\}$. Then, the FSFE module introduces the feature separation factor d_j, defined as the average spectral distance between the current band and the set of bands with greater initial importance ρ_j on the input underwater HSI X. In underwater scenarios, these distances reflect the independence of features across spectral bands. For the ith band, the feature separation factor is calculated as

$$d_i = \frac{1}{i-1} \sum_{k=1}^{i-1} D(X_{\pi(i)}, X_{\pi(k)}). \tag{17}$$

The feature distance $D(X_{\pi(i)}, X_{\pi(k)})$ is calculated using Euclidean distance, which represents the difference between the band $X_{\pi(i)}$ and band $X_{\pi(k)}$

$$D(X_{\pi(i)}, X_{\pi(k)}) = \|X_{\pi(i)} - X_{\pi(k)}\|_2. \tag{18}$$

Introducing the feature separation factor effectively reduces information redundancy caused by high interband correlation, mainly when processing weak signal bands. Subsequently, the feature separation factors of all bands are normalized to yield the normalized feature separation factor $\widehat{d}_i$. To combine the initial importance and independence of the bands, the FSFE module uses the product of the normalized feature separation factor $\widehat{d}_i$ and the initial importance ρ_j of underwater spectral bands to generate the comprehensive evaluation factor δ_j

$$\delta_j = \widehat{d}_i \times \rho_i. \tag{19}$$

The comprehensive evaluation factor δ_j retains band significance information. It reduces the weights of bands that are highly similar to other bands, making it more effective for screening bands with high redundancy in underwater spectral data. Finally, weighted band significance values for the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

underwater HSI X_{BS} are obtained based on the comprehensive evaluation factor $\{\delta_i\}_{i=1}^{C}$ (denoted as $\boldsymbol{\delta}$)

$$X_{\mathrm{BS}} = \boldsymbol{\delta} \odot X = \{\delta_i \odot X_i\}_{i=1}^{C}. \tag{20}$$

Additionally, we index the B key bands ranked highest in the comprehensive evaluation factor as the optimal subset of bands. The selected band number B will be discussed in the experimental analysis section. The optimal subset of bands accurately reflects the important spectral information of underwater HSIs. By combining digital capsules with feature separation factors, our proposed FSFE module significantly improves the accuracy and robustness of underwater spectral profile representation, ensures the representativeness and representativeness of key underwater spectral bands, and reduces redundancy in the spectral data.

D. KBFR Module

The KBFR module is designed to recover the full-band information of the input spectrum by leveraging feature inversion and upsampling operations. It utilizes the reconstructed latent variables output from the capsule network to guide the reconstruction process. By effectively capturing and utilizing the feature information of key bands, KBFR ensures that the original full-band spectrum is accurately restored while preserving essential spectral characteristics. This process enhances the representativeness of the selected bands, ensuring that the optimal band subset retains sufficient spectral information for accurate reconstruction.

The reconstructed latent variables Z' are extracted as high-level features by the DFEB blocks, and the full-band information of the original spectra is progressively recovered using up-sampling operations

$$F_{\mathrm{up}} = \mathrm{Up}(\mathrm{DFEB}(Z')) \tag{21}$$

where F_{up} denotes the features after inversion and upsampling, and $\mathrm{Up}(\cdot)$ denotes the upsampling operation based on the inverse convolution operation and activation function.

The extracted features F_{up} are then fed into the next DFEB block, which uses a 1×1 convolution operation to generate the final reconstructed input spectrum $\widehat{X}$

$$\widehat{X} = \mathrm{Conv}(\mathrm{DFEB}(F_{\mathrm{up}})). \tag{22}$$

Through dynamic feature extraction and inversion techniques, the output spectrum of the KBFR module integrates the importance factors generated by the capsule network with the feature information of the key bands, thereby reconstructing the latent variables Z' to the original input spectrum. This process relies heavily on key band features, which not only enhance the restoration ability of underwater hyperspectral signals but also fully validate the central role of the comprehensive evaluation factor in the reconstruction task.

E. Loss Function

To achieve efficient reconstruction and band selection of underwater HSIs, a multiobjective joint optimization loss function is designed. This function consists of two parts: image reconstruction loss and band selection loss, which are optimized for spectral reconstruction accuracy and key band selection ability, respectively. For the reconstructed hyperspectral image output by the KBFR module, the optimization objective is to make it as close as possible to the original hyperspectral image. This ensures that the selected band subset effectively represents the original spectral information with high fidelity. Additionally, for the optimal band subset selected by the FSFE module, a band characterization constraint is applied to ensure that the selected bands contain rich information while maintaining low redundancy. During network training, both components are jointly optimized, enabling end-to-end optimization through the designed hybrid loss function.

The image reconstruction loss function L_{rec} evaluates the similarity between the reconstructed spectra $\widehat{X}$ and the input spectra X. It consists of two components: magnitude loss and spectral angular distance loss. The magnitude loss uses the mean squared error as the evaluation metric, while the spectral angular distance loss measures the similarity of spectral shapes. The overall image reconstruction loss is computed by combining these two components, expressed as

$$\begin{aligned} L_{\mathrm{rec}} &= L_{\mathrm{MSE}} + L_{\mathrm{SA}} \\ &= \frac{1}{N}\sum_{i=1}^{N} \|X_i - \widehat{X_i}\|_2^2 + \frac{1}{N}\sum_{i=1}^{N} \arccos\left(\frac{X_i \cdot \widehat{X_i}}{\|X_i\|\|\widehat{X_i}\|}\right) \end{aligned} \tag{23}$$

where N is the number of samples and X_i and $\widehat{X_i}$ are the original and reconstructed spectra of the ith sample, respectively.

The band selection loss function L_{bs} is calculated by weighting the band importance of weighted underwater HSI X_{BS}. It combines variance, IE, and sparsity to select an optimal subset of bands with low redundancy and high significance. The variance loss maximizes the variance of weighted underwater HSIs to improve the discriminative ability of the bands. The IE loss quantifies the amount of information between bands by calculating the entropy value, a key index for measuring the uncertainty in the distribution of band features. The sparsity loss constrains the sparsity of the comprehensive evaluation factor $\boldsymbol{\delta}$ and encourages the selection of fewer but more critical bands. The band selection loss function fuses the above three components through weighting

$$\begin{aligned} L_{\mathrm{bs}} &= L_{\mathrm{var}} + L_H + L_{\mathrm{spars}} \\ &= 1 \Big/ \left(\frac{1}{N}\sum_{i=1}^{N} \|X_{\mathrm{BS}_i} - \overline{X}_{\mathrm{BS}_i}\|_2^2\right) \\ &\quad + 1 \Big/ \left(-\sum_{i=1}^{n} p_i \log(p_i)\right) + \sum_{i=1}^{C} |\delta_i| \end{aligned} \tag{24}$$

where $\overline{X}_{\mathrm{BS}_i}$ denotes the mean value of the weighted underwater HSIon the ith band. Finally, the image reconstruction loss and the band selection loss are weighted and combined to form a comprehensive optimization objective

$$L_{\mathrm{total}} = \alpha L_{\mathrm{rec}} + L_{\mathrm{bs}} \tag{25}$$

where α is the hyperparameter that regulates the balance between the image reconstruction loss and the band selection loss, through this design, the loss function not only minimizes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

spectral reconstruction error but also effectively selects the key bands in underwater HSIs, thereby improving the accuracy of both image reconstruction and band selection.

IV. Experimental Results and Analysis

In this section, we will validate the effectiveness and associated analysis of the band selection method for underwater HSIs based on a capsule network, denoted as CapUBS. We describe the experimental setup, including the experimental dataset, evaluation metrics, and experimental environment. We also conduct the hyperparameter sensitivity analysis. Then, we analyze the effectiveness of the CapUBS module through hyperspectral band selection analysis and classification analysis. Finally, we conduct the ablation experiment to evaluate the contribution of each module.

A. Experimental Datasets

To comprehensively evaluate the proposed methodology's performance and ensure the credibility of the experimental results, this study utilizes the classical hyperspectral datasets from Indian Pines and Pavia University, along with our experimentally collected hyperspectral underwater target datasets InPool and OutPool.

1) Indian Pines: The dataset was acquired using NASA's Airborne Visible and Infrared Imaging Spectrometer over the Indian Pine Barrens near Purdue University, Indiana, USA. The raw data includes 200 spectral bands, with 20 bands removed due to water vapor absorption, leaving bands spanning the electromagnetic spectrum from 0.4 to 2.5 μm. The data feature a spatial resolution of 20 m per pixel, an image size of 145 $\times$ 145 pixels, and 16 classes. The dataset is available at https://www.ehu.eus/ccwintco/index.php?title = Hyperspectral_Remote_Sensing_Scenes

2) Pavia University: The dataset was acquired using the ROSIS imaging spectrometer in Pavia, Italy. It includes 103 active spectral bands spanning the visible and near-infrared range from 0.43 to 0.86 μm. The raw dataset initially contained 103 bands, but 12 bands were excluded due to noise and redundancy. The dataset features a spatial resolution of 1.3 m per pixel, an image size of 610 $\times$ 340 pixels, and nine classes. It is available at https://www.ehu.eus/ccwintco/index.php?title = Hyperspectral_Remote_Sensing_Scenes

3) InPool: The dataset was acquired using a Gaia Field portable spectral imager with a spectral resolution of 2.8 nm and a spectral range of 400–1000 nm in an anechoic pool at the School of Marine Science And Technology, Northwestern Polytechnical University, China. During data preprocessing, 120 valid spectral bands were selected within the 420–700-nm range. The images are 200 $\times$ 200 pixels, with underwater targets made of three different materials: iron, wood, and stone. The dataset is available at https://github.com/lq132069/CapUBS/blob/main/dataset/InPool.mat

4) OutPool: To further validate the applicability of the proposed method in a real aquatic environment, underwater multitarget HSIs were collected in an outdoor swimming pool at Northwestern Polytechnical University, China. During the experiment, a UAV captured HSIs at flight altitudes of 20, 50, and 100 m, providing data from various viewpoints and altitudes to enhance the analysis of the underwater target's spectral properties. The dataset contains 120 spectral bands selected from the 420–700-nm range, with an image size of 140 $\times$ 40 pixels. It is available at https://github.com/lq132069/CapUBS/blob/main/dataset/OutPool.mat

B. Experimental Setting

1) Compared Method: Eight representative band selection methods were selected for comparative experiments to evaluate the performance of our proposed method comprehensively. These methods encompass various strategies and theories, including FNGBS [12] and MEV-SFS [14] based on grouping approaches, SSANet [33] and HLFC [11] based on learning strategies, frameworks such as NC-OC-IE, NC-OC-MVPCA, and TRC-OC-FDPC [25] derived from clustering optimization, and S^4P [13] based on sparse subspace theory.

2) Evaluation Metrics: This experiment categorizes the evaluation metrics into subjective and objective assessments. Subjective evaluation involves comparing classification result graphs from different methods on each dataset with ground-truth labeled graphs to assess classification performance intuitively. Objective evaluation relies on mathematical calculations, using quantitative indicators to provide a more precise assessment. To ensure a comprehensive and reliable quantification of classification accuracy, three commonly used evaluation metrics are employed in this article: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa).

3) Environment Setting: The experiments were performed on a hardware setup consisting of an Intel Core i9-12900KF CPU, GeForce RTX 3080 Ti GPU, 64 GB RAM, and the Windows 11 operating system. The code was implemented using Python 3.9 and the PyTorch 1.12.1 deep learning framework. During training, 20% of the dataset was used, with a batch size of 16 and 100 training batches. The learning rate was set to 0.0001, and the Adam optimizer was employed.

C. Hyperparameter Sensitivity Analysis

This section analyzes two critical hyperparameters of the model: the input cube data size P and the loss function weight α. To fully leverage hyperspectral data's spatial and spectral information, we extracted cube data blocks from HSIs. In the experiments, we analyzed cube sizes P set as 4, 6, 8, 10, 12, and 15. Our proposed hybrid loss function includes image reconstruction loss and band selection loss, combined using a weighting parameter α to balance their contributions during optimization. The α values of 0.1, 1, 10, 100, 1000, and 10 000 were tested and analyzed in the experiments. Experiments were conducted on four datasets to analyze the impact of hyperparameters P and α, using OA, AA, and Kappa as evaluation metrics. The results are presented in Fig. 4.

From the experimental results, the performance of our proposed CapUBS varies across datasets under different hyperparameter configurations. For the Indian Pines dataset, model performance increases initially and decreases as P increases,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

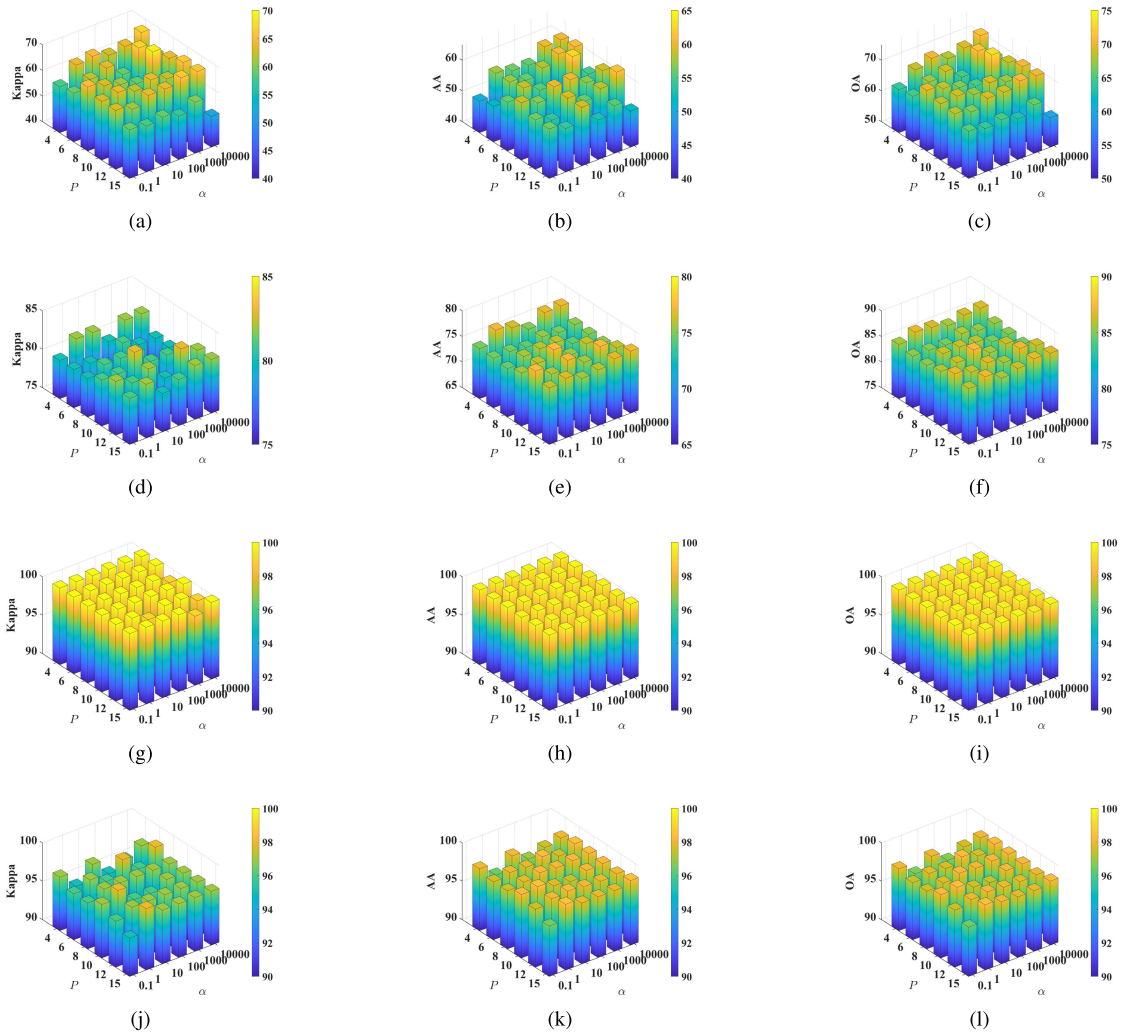


Fig. 4. Plots of the hyperparameter analysis results for the proposed model on four datasets. (a)–(c) On the Indian Pines dataset, evaluated using Kappa, AA, and OA. (d)–(f) On the Pavia University dataset, with Kappa, AA, and OA metrics. (g)–(i) On the InPool dataset, using Kappa, AA, and OA. (j)–(l) On the OutPool dataset, with Kappa, AA, and OA evaluation metrics. The best hyperparameter configuration for each dataset is adopted in subsequent experiments to ensure optimal model performance.

with significant improvement observed when α exceeds 10. Performance stabilizes at $\alpha \geq 1000$, achieving the best performance at $P = 8$ and $\alpha = 1000$. On the Pavia University dataset, performance shows less variation with changes in P and α. The best band selection occurs at $P = 10$ and $\alpha = 10$, optimizing the subsequent classification accuracy. For the InPool dataset, overall classification accuracy is higher with less fluctuation, and the optimal configuration is $P = 6$ and $\alpha = 1000$. The OutPool dataset exhibits lower OA with more significant fluctuation, performing best at $P = 10$ and $\alpha = 1$. Based on these analyses, the best hyperparameter configuration for each dataset is adopted in subsequent experiments to ensure optimal model performance.

D. Band Number Analysis

This section analyzes the effect of selecting different band numbers on each dataset's subsequent hyperspectral classification performance. It also compares the performance of our proposed CapUBS with other methods using OA, AA, and Kappa, as shown in Fig. 5. The experimental results demonstrate that as the number of bands increases, the classification performance across all four datasets generally improves, though the extent of improvement varies. For the Indian Pines dataset in Fig. 5(a)–(c), both OA and AA metrics steadily increase with the number of bands, while the Kappa coefficient also shows improvement after 21 bands. Considering the redundancy of bands and performance balance, 21 bands were selected for further analysis. For the Pavia University dataset in Fig. 5(d)–(f), OA, AA, and Kappa significantly increased when the number of bands was below 21, with improvements leveling off afterward, suggesting that the addition of redundant information no longer contributes to better performance.

For the InPool dataset in Fig. 5(g)–(i), the classification performance stabilizes after 21 bands, with AA and OA indexes remaining relatively unchanged. The Kappa of our proposed method shows minimal improvement once the number of bands exceeds 21, indicating that 21 bands are sufficient to represent the data. For the OutPool dataset in Fig. 5(j)–(l), the classification accuracy curve fluctuates significantly, but overall, accuracy tends to improve with an increasing number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

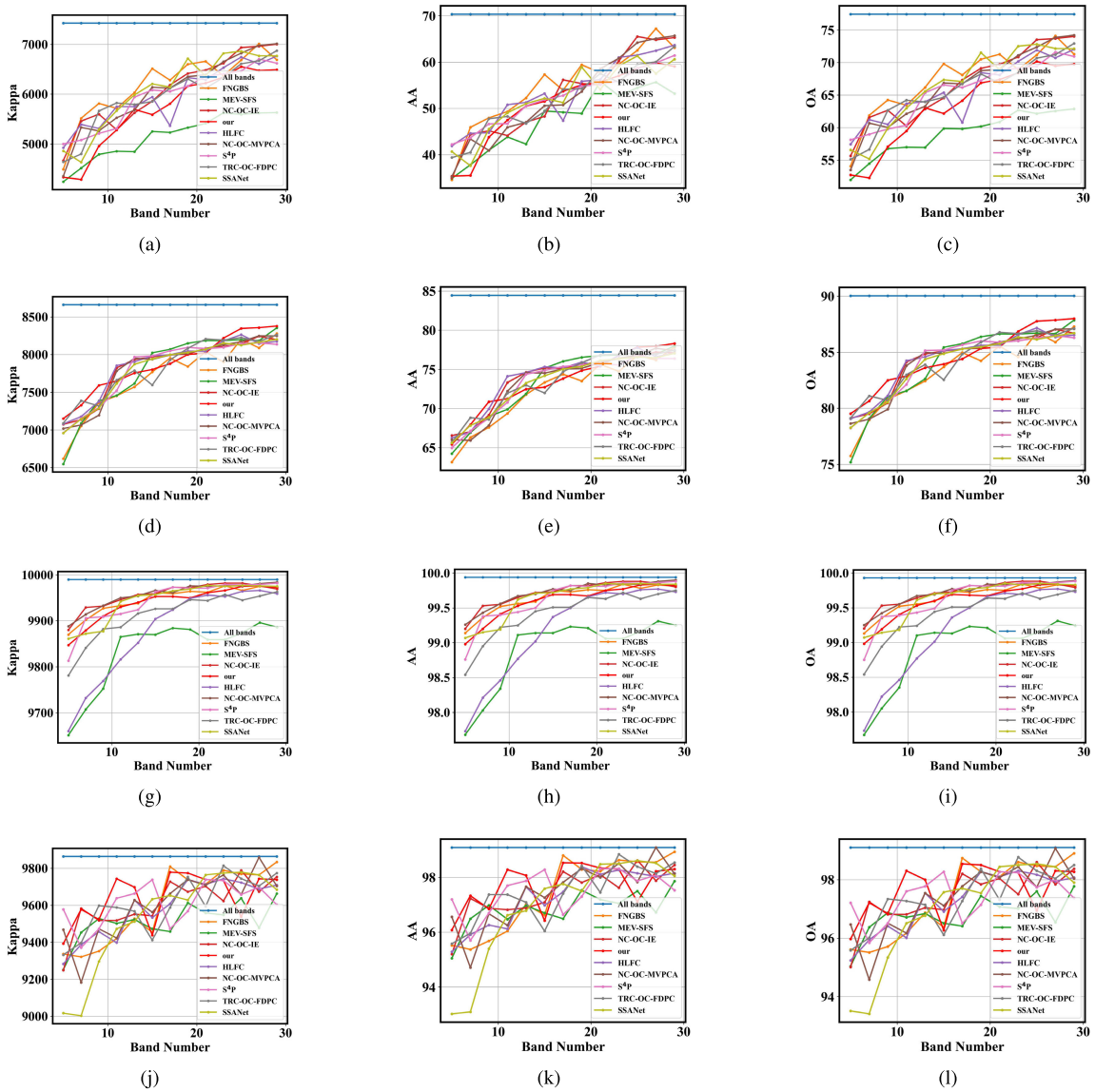


Fig. 5. Classification performance of band selection methods on different band numbers. (a)–(c) Kappa, AA, and OA evaluation metrics on the Indian Pines dataset. (d)–(f) Kappa, AA, and OA evaluation metrics on the Pavia University dataset. (g)–(i) Kappa, AA, and OA evaluation metrics on the InPool dataset. (j)–(l) Kappa, AA, and OA evaluation metrics on the OutPool dataset. 21 bands were consistently selected across all datasets for the hyperspectral classification analysis.

of bands. When the number of bands is too low, the feature characterization is insufficient, while too many bands introduce redundancy. As a result, 21 bands were also chosen for subsequent analysis. In summary, 21 bands were consistently selected across all datasets for the hyperspectral classification analysis to maintain classification accuracy while controlling band redundancy.

E. Band Selection and Classification Results

1) Results on the Indian Pines Dataset: We compare the differences in band selection between the proposed method and the other methods on the Indian Pines dataset. Fig. 6(a) visualizes the distribution of the selected bands for each method, while Fig. 6(b) demonstrates the entropy values for each band. The bands selected by our proposed method exhibit a broader range, a more uniform distribution, and better concentration, allowing for a more comprehensive dataset characterization.

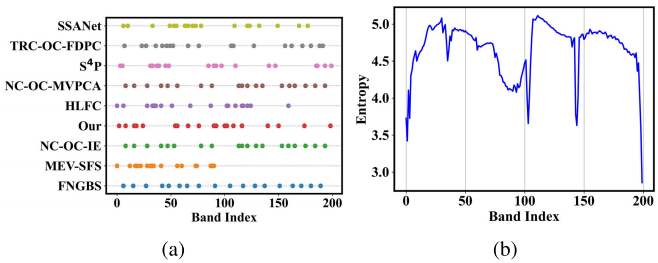


Fig. 6. (a) Distribution of the selected band subsets for each band selection method on the Indian Pines dataset. (b) Entropy values for each spectral band.

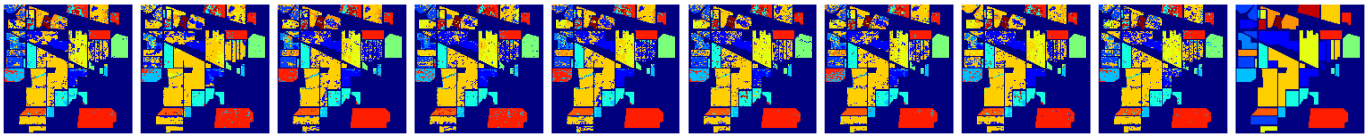
In contrast, the band distributions of TRC-OC-FDPC and S^4P are overly dispersed, making it challenging to capture the complete characteristics of the data. On the other hand, the band selections of MEV-SFS and HLFC are overly centralized and limited in scope, failing to represent the dataset's overall

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

TABLE I

CLASSIFICATION ACCURACY OF EACH BAND SELECTION METHOD USING 21 BANDS ON THE INDIAN PINES DATASET. THE BEST VALUE IS HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST VALUE IS INDICATED IN BLUE

NO.	All band	FNGBS	MEV-SFS	NC-OC-IE	HLFC	NC-OC-MVPCA	S^4P	TRC-OC-FDPC	SSANet	Our
OA	77.14±0.7	71.22±1.05	60.47±0.55	70.24±0.6	67.4±1.05	69.31±0.87	69.27±0.82	69.73±1.12	64.08±1.7	**72.31±0.7**
AA	72.96±1.19	60.19±2.08	51.87±1.45	59.7±1.5	56.96±1.92	57.04±2.53	56.66±1.34	59.62±1.95	54.87±2.1	**61.05±1.38**
Kappa	73.84±0.8	66.64±1.33	53.38±0.7	65.46±0.79	61.93±1.32	64.33±1.09	64.32±1.04	64.83±1.44	58.04±2.27	**67.88±0.87**
1	56.25±8.14	47.18±31.48	35.48±21.08	54.82±35.6	35.92±16.86	21.71±25.58	49.05±13.94	**81.33±19.21**	43.7±19.41	27.0±28.96
2	72.62±1.8	66.19±2.27	47.83±2.24	67.12±3.62	61.11±3.58	66.71±2.86	62.9±2.41	65.48±2.77	58.45±3.27	**71.37±1.99**
3	68.38±2.11	66.02±3.65	47.1±10.37	**77.83±3.61**	62.6±7.19	77.69±3.97	68.65±2.35	66.41±5.03	56.8±4.22	73.05±2.64
4	65.79±5.41	49.03±5.26	46.09±5.25	55.28±6.71	60.06±6.84	55.53±10.95	61.7±10.04	64.64±5.5	54.61±6.2	**70.35±5.59**
5	85.71±3.49	72.6±3.3	**80.08±3.88**	63.69±5.75	78.45±2.53	57.65±7.35	76.52±3.34	73.42±3.37	71.44±3.85	75.95±4.54
6	91.25±1.86	**87.28±1.71**	83.71±2.14	79.99±3.02	85.5±2.78	81.83±2.6	82.47±3.15	82.2±1.5	86.21±2.13	83.78±2.76
7	90.29±6.36	**88.85±9.66**	67.41±7.9	85.7±14.56	68.8±13.15	81.79±12.87	43.3±38.45	79.98±17.25	44.75±14.83	82.26±13.98
8	94.19±0.94	89.49±1.69	90.46±1.53	**91.01±1.86**	90.01±1.75	89.15±1.67	87.36±1.54	90.35±1.78	90.88±1.84	88.53±1.39
9	74.35±19.18	0.0±0.0	20.39±20.28	0.0±0.0	26.25±34.39	6.67±22.61	35.5±42.36	15.0±35.71	10.42±26.99	**46.46±47.53**
10	74.13±2.86	**73.6±2.46**	42.95±4.58	72.88±2.63	64.69±3.92	70.88±3.32	64.11±2.83	70.28±4.52	46.94±5.61	67.24±2.73
11	74.44±1.19	62.07±3.0	48.88±1.32	61.93±2.21	55.33±2.17	61.11±1.67	60.95±3.06	60.54±2.58	53.81±4.09	**62.93±1.88**
12	71.15±4.32	**55.02±9.27**	44.39±17.49	53.51±6.19	45.57±13.04	54.12±5.17	50.74±6.44	52.55±6.82	51.7±8.23	46.37±5.77
13	93.07±3.27	86.02±2.96	87.13±2.99	86.43±4.87	84.34±3.18	87.46±3.19	87.75±3.54	81.19±7.08	82.94±3.38	**88.5±3.49**
14	93.13±0.97	88.21±1.21	88.07±1.48	80.75±3.4	89.26±1.49	79.41±3.87	87.69±1.32	86.13±1.22	88.06±1.43	**89.31±1.23**
15	75.75±3.03	**82.14±3.72**	56.68±3.13	77.07±3.8	71.59±4.62	72.23±5.79	70.42±3.77	62.93±4.47	66.15±4.8	78.29±3.06
16	97.41±2.44	99.09±0.85	99.58±0.57	98.67±1.95	99.32±1.05	98.89±1.63	97.27±1.92	99.65±0.82	**99.82±0.58**	98.19±2.96



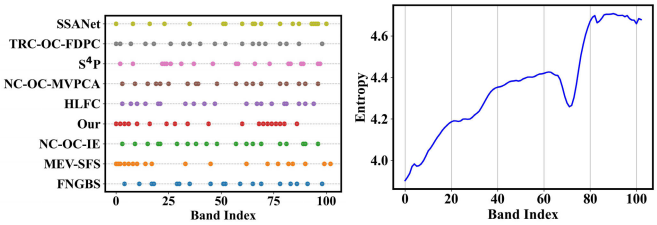
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 7. SVM classification results for each band selection method using 21 bands on the Indian Pines dataset. (a) FNGBS. (b) MEV-SFS. (c) NC-OC-IE. (d) HLFC. (e) NC-OC-MVPCA. (f) S^4P. (g) TRC-OC-FDPC. (h) SSANet. (i) Ours. (j) Ground truth.

features fully. Additionally, Fig. 6(b) reveals that the entropy value of the proposed method is higher, indicating that it selects more important bands while reducing redundancy. This further highlights the superiority of the proposed method in band selection.

To visually and quantitatively highlight the advantages of the proposed method, we compare the classification accuracies of different band selection methods on the Indian Pines dataset. The classification results of the nine-band selection methods, based on the selected 21 bands, are visualized in Fig. 7, while Table I provides the corresponding classification accuracies in quantitative form. The experimental results demonstrate that the method proposed in this article outperforms the other eight methods in terms of OA, AA, and Kappa. Furthermore, across the 16 classification categories, our proposed method achieves the best performance in 6 categories and ranks second in 1 category, showcasing its superior overall performance compared to the other methods.

2) Results on the Pavia University Dataset: Fig. 8(a) visualizes the distribution of the selected bands across different band selection methods on the Pavia University dataset. The proposed method in this article exhibits a denser distribution in the 0–20th and 60–80th band intervals, reflecting a higher degree of selectivity. Compared to other band selection methods, our method demonstrates superior aggregation, outperforming methods like SSANet, TRC-OC-FDPC, and NC-OC-IE while showing a more uniform distribution than



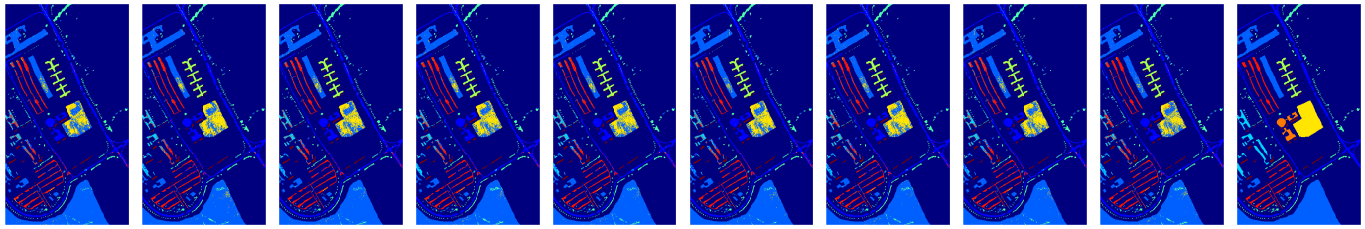
(a) (b)

Fig. 8. (a) Distribution of the selected band subsets for each band selection method on the Pavia University dataset. (b) Entropy values for each spectral band.

MEV-SFS. In Fig. 8(b), the entropy analysis indicates that the proposed method achieves higher entropy values, suggesting the selection of more important bands with reduced redundancy, further underscoring its superiority.

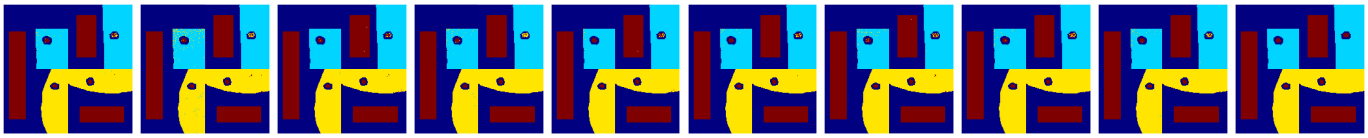
Fig. 9 presents the visualized classification results of different methods using the selected bands on the Pavia University dataset, while Table II provides the quantitative classification accuracies. The experimental results demonstrate that the proposed method significantly outperforms the other comparative methods in terms of OA, AA, and Kappa coefficient, achieving optimal performance. Our proposed method excels in classification accuracy in 4 out of 9 categories and performs second best in two categories, showing a clear advantage over the other methods. Fig. 10 presents the intuitive results of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. SVM classification results for each band selection method using 21 bands on the Pavia University dataset. (a) FNGBS. (b) MEV-SFS. (c) NC-OC-IE. (d) HLFC. (e) NC-OC-MVPCA. (f) S^{4}P. (g) TRC-OC-FDPC. (h) SSANet. (i) Ours. (j) Ground truth.



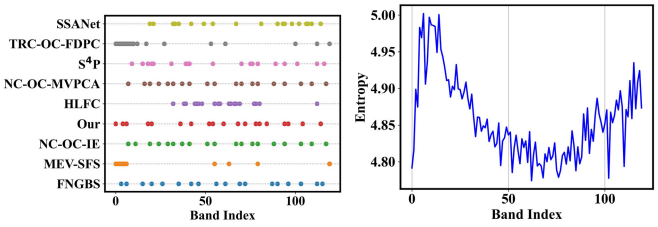
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 10. SVM classification results for each band selection method using 21 bands on the InPool dataset. (a) FNGBS. (b) MEV-SFS. (c) NC-OC-IE. (d) HLFC. (e) NC-OC-MVPCA. (f) S^{4}P. (g) TRC-OC-FDPC. (h) SSANet. (i) Ours. (j) Ground truth.

TABLE II

CLASSIFICATION ACCURACY OF EACH BAND SELECTION METHOD USING 21 BANDS ON THE PAVIA UNIVERSITY DATASET. THE BEST VALUE IS HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST VALUE IS INDICATED IN BLUE

NO.	All band	FNGBS	MEV-SFS	NC-OC-IE	HLFC	NC-OC-MVPCA	S^{4}P	TRC-OC-FDPC	SSANet	Our
OA	90.14±0.22	86.33±0.13	86.36±0.25	85.68±0.17	85.78±0.15	85.83±0.16	85.91±0.22	86.49±0.16	82.52±0.34	**86.95±0.18**
AA	84.57±0.89	76.57±0.24	76.52±0.42	75.78±0.3	76.03±0.26	75.93±0.26	75.74±0.34	76.51±0.24	70.88±0.57	**77.25±0.33**
Kappa	86.74±0.3	81.43±0.18	81.51±0.35	80.51±0.24	80.69±0.21	80.73±0.23	80.85±0.29	81.66±0.21	75.99±0.5	**82.32±0.25**
1	86.93±1.28	79.14±0.31	79.62±0.31	79.43±0.31	79.16±0.36	79.16±0.22	78.16±0.43	79.29±0.4	78.0±0.5	**79.76±0.31**
2	91.72±0.32	88.09±0.25	88.62±0.32	87.34±0.33	87.76±0.32	87.69±0.33	88.06±0.24	88.61±0.29	84.19±0.52	**89.24±0.33**
3	84.34±0.95	**83.64±1.51**	82.87±2.22	81.48±1.61	82.3±1.72	81.08±1.89	81.98±1.9	82.23±2.1	68.66±1.23	82.91±1.94
4	95.01±1.17	91.09±1.55	90.38±1.5	91.75±1.04	91.02±1.28	92.12±1.39	91.37±1.74	91.54±1.08	89.81±1.36	**92.56±1.25**
5	99.42±0.39	99.44±0.32	99.49±0.36	99.67±0.24	99.43±0.49	99.24±0.39	99.57±0.3	99.61±0.22	99.71±0.22	**99.72±0.19**
6	90.0±1.14	88.82±1.16	87.56±0.77	87.36±1.75	86.42±1.82	87.66±1.38	**88.91±0.92**	88.62±1.16	85.66±1.32	87.62±1.52
7	86.75±1.93	0.0±0.0	1.28±3.18	0.56±2.42	**5.64±13.76**	4.02±8.98	0.93±3.21	0.36±1.56	0.83±3.63	0.0±0.0
8	84.12±0.81	**80.33±1.0**	79.17±1.23	78.45±0.97	79.19±0.69	78.45±1.03	78.36±1.04	79.57±1.2	71.78±1.6	79.91±0.97
9	99.97±0.05	99.94±0.07	99.96±0.05	99.94±0.06	99.98±0.05	**99.98±0.04**	99.96±0.05	99.95±0.06	**99.98±0.04**	99.96±0.05



(a) (b)

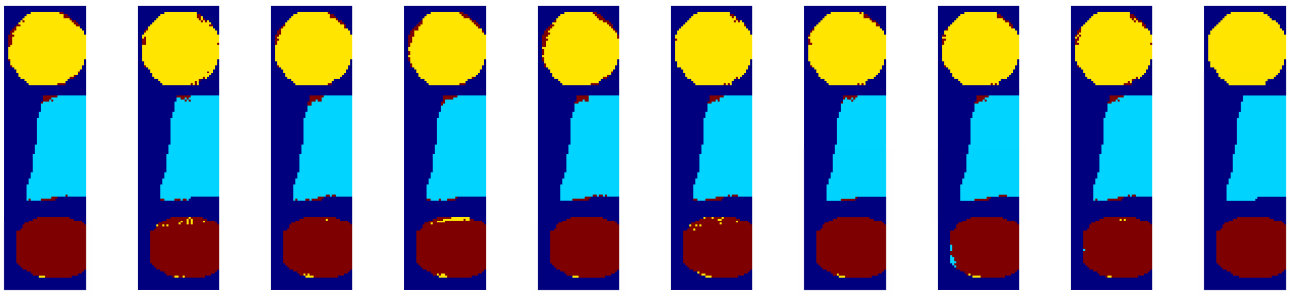
Fig. 11. (a) Distribution of the selected band subsets for each band selection method on the InPool dataset. (b) Entropy values for each spectral band.

different models for classifying the selected bands on the InPool dataset, while Table III provides the corresponding classification accuracies.

3) Results on the InPool Dataset: Fig. 11(a) visualizes the comparison of the distribution of selected bands on the InPool dataset. The results show that our proposed method exhibits a more uniform distribution of selected bands, which is significantly better than TRC-OC-FDPC, HLFC, and MEV-SFS, whose band distributions are overly concentrated and fail to characterize the data fully. Compared to methods like SSANet and S^{4}P, our proposed method demonstrates more substantial aggregation and selects important bands in a more targeted manner. This makes it particularly suitable for scenarios with clear mission objectives while also maintaining a degree of generality. Additionally, Fig. 11(b) shows that the entropy value of the selected bands in our proposed method is significantly higher than that of the other methods. This indicates that the proposed method effectively selects representative and important bands while reducing redundancy, emphasizing its advantages in the band selection task.

Our proposed method outperforms the other compared methods in terms of overall performance. This method achieves the best results in OA, AA, and Kappa. The method also performs well, achieving optimal classification accuracy for Categories 2 and 3 while maintaining superior overall performance compared to the other methods. Table IV presents the corresponding classification accuracies. Fig. 12 shows the intuitive results of different models for classifying the selected bands on the OutPool dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.



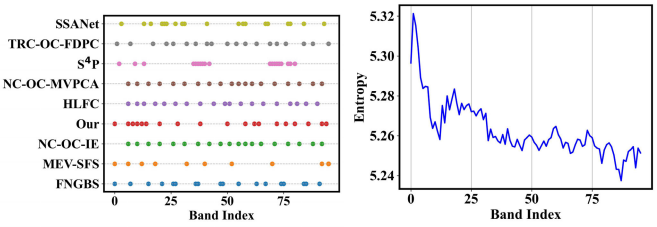
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 12. SVM classification results for each band selection method using 21 bands on the OutPool dataset. (a) FNGBS. (b) MEV-SFS. (c) NC-OC-IE. (d) HLFC. (e) NC-OC-MVPCA. (f) S^4P. (g) TRC-OC-FDPC. (h) SSANet. (i) Ours. (j) Ground truth.

TABLE III

CLASSIFICATION ACCURACIES OF EACH BAND SELECTION METHOD USING 21 BANDS ON THE INPOOL DATASET. THE BEST VALUE IS HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST VALUE IS INDICATED IN BLUE

NO.	All band	FNGBS	MEV-SFS	NC-OC-IE	HLFC	NC-OC-MVPCA	S^4P	TRC-OC-FDPC	SSANet	Our
OA	99.9±0.05	99.76±0.04	99.18±0.06	**99.83±0.03**	99.7±0.07	99.62±0.06	99.81±0.06	99.62±0.05	99.69±0.07	**99.83±0.03**
AA	99.9±0.05	99.76±0.04	99.18±0.06	99.83±0.04	99.71±0.06	99.62±0.06	99.81±0.06	99.62±0.05	99.69±0.07	**99.83±0.03**
Kappa	99.85±0.07	99.64±0.06	98.76±0.08	99.74±0.05	99.56±0.1	99.43±0.09	99.71±0.1	99.42±0.07	99.53±0.1	**99.75±0.04**
1	99.96±0.03	99.94±0.04	99.59±0.18	**99.96±0.01**	99.95±0.02	99.85±0.07	99.96±0.03	99.87±0.06	99.96±0.02	99.96±0.04
2	99.76±0.11	99.44±0.12	98.05±0.16	99.59±0.07	99.23±0.21	99.09±0.17	99.55±0.18	99.14±0.14	99.17±0.2	**99.6±0.08**
3	99.99±0.01	99.93±0.04	99.9±0.06	99.95±0.03	99.96±0.04	**99.96±0.03**	99.96±0.06	99.9±0.04	99.94±0.04	**99.96±0.03**



(a) (b)

Fig. 13. (a) Distribution of the selected band subsets for each band selection method on the OutPool dataset. (b) Entropy values for each spectral band.

4) Results on the OutPool Dataset: For the OutPool dataset, Fig. 13(a) visually demonstrates the differences in band selection between the proposed and other comparative methods. The results show that the bands selected by our proposed method are more uniformly distributed, covering the data features more comprehensively than those of SSANet, S^4P, and MEV-SFS. Furthermore, compared with HLFC, NC-OC-IE, and TRC-OC-FDPC, the proposed method is more aggregated in specific bands, making it more suitable for targeted classification tasks. Furthermore, Fig. 13(b) shows that the entropy value of the selected bands for the proposed method is significantly important. This indicates that our proposed method effectively selects important bands crucial for feature characterization while reducing redundancy, providing more substantial support for task classification. This performance further reinforces the advantage of the proposed method in band selection.

The combined performance of our CapUBS significantly outperforms the comparison methods, achieving outstanding results in all metrics. The OA and Kappa coefficient reach their optimal values, while the AA is slightly inferior but still close to the suboptimal. Additionally, the model achieves the best performance in category 3, with classification accuracy for the other two categories close to the optimal, ensuring that the overall performance remains superior.

The results presented in Tables I–IV show that our proposed method achieves consistently high performance across the experimental datasets. To further investigate its applicability in underwater environments, we analyzed the spectral distribution of the selected band subsets. As illustrated in Figs. 6, 8, 11, and 13, the bands selected by our method are predominantly concentrated within the blue-green spectral region (blue: 450–495 nm, green: 495–570 nm). This distribution is consistent with the optical properties of water, where blue and green light exhibit superior penetration. In comparison, the band subsets identified by other methods tend to be more dispersed or biased toward spectral regions with limited underwater transmittance. It suggests that our method effectively captures spectral characteristics that are particularly relevant for underwater scenarios.

F. Ablation Experiment

To investigate the role of each module in improving the overall performance of our proposed method and assess whether each module contributes positively, we have designed an ablation experiment.

We design four model variants to conduct a detailed ablation analysis. In the ablation experiments, for the DLVE module, the variants retain only the original spectral feature extraction branch while removing the guide effect of the average spectral features. The experimental results further validate the importance of each module by comparing the performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

CLASSIFICATION ACCURACIES OF EACH BAND SELECTION METHOD USING 21 BANDS ON THE OUTPOOL DATASET. THE BEST VALUE IS HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST VALUE IS INDICATED IN BLUE

NO.	All band	FNGBS	MEV-SFS	NC-OC-IE	HLFC	NC-OC-MVPCA	S^4P	TRC-OC-FDPC	SSANet	Our
OA	98.83±0.29	97.92±0.54	97.15±0.64	97.87±0.45	97.82±0.6	97.87±0.5	97.79±0.37	98.08±0.41	**98.32±0.32**	98.11±0.25
AA	98.84±0.27	97.97±0.54	97.2±0.69	97.92±0.45	97.86±0.62	97.93±0.49	97.86±0.4	98.16±0.4	**98.39±0.31**	98.15±0.27
Kappa	98.22±0.44	96.85±0.81	95.7±0.96	96.79±0.69	96.7±0.91	96.78±0.76	96.66±0.57	97.1±0.62	96.46±0.48	**97.14±0.38**
1	99.85±0.28	**99.98±0.05**	99.78±0.51	99.79±0.47	99.92±0.36	99.95±0.1	99.85±0.4	99.95±0.17	99.86±0.34	99.92±0.13
2	99.41±0.28	98.99±0.95	98.86±1.14	99.09±0.73	98.99±0.92	99.14±0.48	99.03±0.58	99.45±0.34	**99.56±0.2**	98.98±0.73
3	96.86±1.01	94.21±1.59	92.04±1.72	94.18±1.47	93.94±1.78	93.83±1.44	93.83±0.87	94.25±1.26	94.31±0.85	**94.83±0.94**

TABLE V

ABLATION ANALYSIS ON FOUR DATASETS. **BOLD** INDICATES THE BEST VALUES AND THE SECOND-BEST VALUE IS INDICATED IN BLUE

Variant Models			Indian Pines			Pavia University			InPool			OutPool		
DLVE	CNBC	FSFE	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
✓	✓	✓	**72.11**	59.7	**67.77**	**87.71**	**78.14**	**83.41**	**99.88**	**99.88**	**99.82**	**98.7**	**98.71**	**98.04**
✗	✓	✓	69.29	60.28	64.18	86.47	75.13	81.69	99.77	99.77	99.65	97.94	98.02	96.88
✗	✓	✗	66.15	54.19	60.45	85.64	73.57	80.52	99.6	99.61	99.4	97.14	97.25	95.68
✓	✗	✓	68.44	**60.57**	63.42	87.01	78.08	83.21	99.81	99.81	99.71	96.87	97.06	95.6
✓	✗	✗	69.63	58.38	64.81	87.57	76.64	82.37	99.58	99.58	99.37	94.91	94.69	92.3

of these variants with the full model, as shown in Table V. Specifically, the FSFE module significantly improves classification performance, suggesting that this module helps the CapUBS generate a more representative and less redundant set of bands. Retaining all modules of the CapUBS resulted in significantly better performance than the four variants, further indicating that the modules were designed to be essential, not redundant. Additionally, the synergy between the DLVE, CNBC, and FSFE modules significantly enhances overall model performance while effectively mitigating underwater spectral offsets and redundancy issues.

V. CONCLUSION

This article proposes a capsule network-based band selection method for underwater HSIs to address spectral offset variation and redundancy in band selection. First, the DLVE module inputs the original and average spectra. It extracts local details and global semantic features through DFEBs to mitigate the underwater spectral variation. Then, the CNBC module generates digital capsules to characterize joint multiband features through a dynamic routing mechanism. It performs multiband feature extraction and quantization, suppresses spectral offset, characterizes the multidimensional attributes of the spectra using vector directions, and quantifies the initial importance of the spectral bands via vector norm. The FSFE module calculates the initial band importance ranking and feature separation factor to identify key bands with low redundancy and high representativeness, thereby obtaining the optimal subset of bands. Finally, the KBFR module recovers the full band spectrum based on the key band features.

The quantitative and visual analysis results on four classical hyperspectral datasets and underwater hyperspectral experimental datasets demonstrate that the effectiveness of our proposed method outperforms. The dynamic routing mechanism of the capsule network and vector feature representation effectively suppress the spectral offset variation caused by distortion in the underwater environment while capturing the complex characteristics of underwater HSIs. Additionally, the comprehensive band evaluation strategy ensures the representativeness and representativeness of the band selection results, providing a more stable and reliable foundation for target detection and classification of underwater HSIs. In future work, we will focus on integrating band selection and other image preprocessing algorithms with downstream tasks to further enhance their applicability in underwater hyperspectral analysis. Additionally, we plan to explore adaptive band selection strategies that dynamically adjust to different underwater environments, improving the generalization and efficiency of our method.

REFERENCES

[1] S. Feng et al., "Transformer-based cross-domain few-shot learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5501716.

[2] Q. Li, J. Li, T. Li, Z. Li, and P. Zhang, "Spectral–spatial depth-based framework for hyperspectral underwater target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4204615.

[3] Q. Li, J. Li, T. Li, and Y. Feng, "A joint framework for underwater hyperspectral image restoration and target detection with conditional diffusion model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 17263–17277, 2024.

[4] L. Deng, Z. Mao, X. Li, Z. Hu, F. Duan, and Y. Yan, "UAV-based multispectral remote sensing for precision agriculture: A comparison between different cameras," *ISPRS J. Photogramm. Remote Sens.*, vol. 146, pp. 124–136, Dec. 2018.

[5] K. K.-K. Kang, M. Hoekstra, M. Foroutan, A. M. Chegoonian, K. Zolfaghari, and C. R. Duguay, "Operating procedures and calibration of a hyperspectral sensor onboard a remotely piloted aircraft system for water and agriculture monitoring," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 9200–9203.

[6] P. Duan, T. Shan, X. Kang, and S. Li, "Spectral super-resolution in frequency domain," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Oct. 29, 2024, doi: 10.1109/TNNLS.2024.3481060.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

[7] S. Jay et al., "Hyperspectral remote sensing of shallow waters: Considering environmental noise and bottom intra-class variability for modeling and inversion of water reflectance," *Remote Sens. Environ.*, vol. 200, pp. 352–367, Oct. 2017.
[8] C. A. Williamson and R. C. Hollins, "Measured IOPs of Jerlov water types," *Appl. Opt.*, vol. 61, no. 33, pp. 9951–9961, 2022.
[9] M. Darwiesh, A. F. El-Sherif, M. F. Hassan, H. S. Ayoub, and Y. H. Elbashar, "Simulation and characterization of underwater target detection using LiDAR system," *J. Opt.*, vol. 49, no. 3, pp. 416–426, Sep. 2020.
[10] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.
[11] J. Wang et al., "Region-aware hierarchical latent feature representation learning-guided clustering for hyperspectral band selection," *IEEE Trans. Cybern.*, vol. 53, no. 8, pp. 5250–5263, Aug. 2022.
[12] Q. Wang, Q. Li, and X. Li, "A fast neighborhood grouping method for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5028–5039, Jun. 2021.
[13] C. Tang et al., "Spatial and spectral structure preserved self-representation for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531413.
[14] W. Zhang, X. Li, Y. Dou, and L. Zhao, "A geometry-based band selection approach for hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4318–4333, Aug. 2018.
[15] A. Rodriguez and A. Laio, "Clustering by fast search and find of density peaks," *Science*, vol. 344, no. 6191, pp. 1492–1496, Jun. 2014.
[16] S. Jia, G. Tang, J. Zhu, and Q. Li, "A novel ranking-based clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 88–102, Jan. 2016.
[17] W. Zhang, X. Li, and L. Zhao, "Band priority index: A feature selection framework for hyperspectral imagery," *Remote Sens.*, vol. 10, no. 7, p. 1095, Jul. 2018.
[18] B. Guo, S. R. Gunn, R. I. Damper, and J. D. B. Nelson, "Band selection for hyperspectral image classification using mutual information," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 4, pp. 522–526, Oct. 2006.
[19] M. Gong, M. Zhang, and Y. Yuan, "Unsupervised band selection based on evolutionary multiobjective optimization for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 544–557, Jan. 2016.
[20] A. Fontanella, E. Marenzi, E. Torti, G. Danese, A. Plaza, and F. Leporati, "A suite of parallel algorithms for efficient band selection from hyperspectral images," *J. Real-Time Image Process.*, vol. 15, no. 3, pp. 537–553, Oct. 2018.
[21] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Nov. 1999.
[22] C.-I. Chang and S. Wang, "Constrained band selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1575–1585, Jun. 2006.
[23] H. Yang, Q. Du, H. Su, and Y. Sheng, "An efficient method for supervised hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 138–142, Jan. 2011.
[24] J. Feng, L. C. Jiao, X. Zhang, and T. Sun, "Hyperspectral band selection based on trivariate mutual information and clonal selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4092–4105, Jul. 2014.
[25] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.
[26] W. Xie, J. Lei, J. Yang, Y. Li, Q. Du, and Z. Li, "Deep latent spectral representation learning-based hyperspectral band selection for target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2015–2026, Mar. 2020.
[27] C. Cheng, M. Wang, K. Wu, W. Liu, Z. Tang, and M. Chen, "DISGT: Dynamic-interactive subgraph transformer for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5536814.
[28] J. Feng et al., "Convolutional neural network based on bandwise-independent convolution and hard thresholding for hyperspectral band selection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4414–4428, Sep. 2021.
[29] P. Duan, S. Hu, X. Kang, and S. Li, "Shadow removal of hyperspectral remote sensing images with multiexposure fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537211.
[30] B. Xi et al., "MCTGCL: Mixed CNN–transformer for Mars hyperspectral image classification with graph contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5503214.
[31] T. Liu et al., "AEKAN: Exploring superpixel-based AutoEncoder Kolmogorov–Arnold network for unsupervised multimodal change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5601114.
[32] T. Liu et al., "Commonality feature representation learning for unsupervised multimodal change detection," *IEEE Trans. Image Process.*, vol. 34, pp. 1219–1233, 2025.
[33] C. Cui, X. Sun, B. Fu, and X. Shang, "SSANet-BS: Spectral–spatial cross-dimensional attention network for hyperspectral band selection," *Remote Sens.*, vol. 16, no. 15, p. 2848, Aug. 2024.
[34] J. Feng et al., "Multi-agent deep reinforcement learning for hyperspectral band selection with hybrid teacher guide," *Knowl.-Based Syst.*, vol. 299, Sep. 2024, Art. no. 112044.
[35] S. O. Atik, "Dual-stream spectral–spatial convolutional neural network for hyperspectral image classification and optimal band selection," *Adv. Space Res.*, vol. 74, no. 5, pp. 2025–2041, Sep. 2024.
[36] H. Zhang, H. Gao, H. Sun, X. Sun, and B. Zhang, "A spatial-spectrum fully-attention network for band selection of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[37] Y. Cai, X. Liu, and Z. Cai, "BS-nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Mar. 2020.
[38] Y. Zhan et al., "A new hyperspectral band selection approach based on convolutional neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 3660–3663.
[39] P. Ribalta Lorenzo, L. Tulczyjew, M. Marcinkiewicz, and J. Nalepa, "Hyperspectral band selection using attention-based convolutional neural networks," *IEEE Access*, vol. 8, pp. 42384–42403, 2020.
[40] Z. Dou, K. Gao, X. Zhang, H. Wang, and L. Han, "Band selection of hyperspectral images using attention-based autoencoders," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 147–151, Jan. 2021.
[41] S. K. Roy, S. Das, T. Song, and B. Chanda, "DARecNet-BS: Unsupervised dual-attention reconstruction network for hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 12, pp. 2152–2156, Dec. 2021.
[42] J. Qi et al., "Bathymetric-based band selection method for hyperspectral underwater target detection," *Remote Sens.*, vol. 13, no. 19, p. 3798, Sep. 2021.
[43] X. Fu, X. Shang, X. Sun, H. Yu, M. Song, and C.-I. Chang, "Underwater hyperspectral target detection with band selection," *Remote Sens.*, vol. 12, no. 7, p. 1056, Mar. 2020.
[44] M. Darwiesh, H. S. Ayoub, A. F. El-Sherif, and Y. H. Elbashar, "Airborne hyperspectral detection of underwater camouflaged targets and the effect of target shape and seafloor on image quality," *J. Opt.*, vol. 50, no. 1, pp. 7–27, Mar. 2021.
[45] M. Babin et al., "Variations in the light absorption coefficients of phytoplankton, nonalgal particles, and dissolved organic matter in coastal waters around Europe," *J. Geophys. Res., Oceans*, vol. 108, no. C7, Jul. 2003.
[46] M. Babin, A. Morel, V. Fournier-Sicre, F. Fell, and D. Stramski, "Light scattering properties of marine particles in coastal and open ocean waters as related to the particle mass concentration," *Limnology Oceanogr.*, vol. 48, no. 2, pp. 843–859, Mar. 2003.
[47] Z. Lee, K. L. Carder, C. D. Mobley, R. G. Steward, and J. S. Patch, "Hyperspectral remote sensing for shallow waters: 2. Deriving bottom depths and water properties by optimization," *Appl. Opt.*, vol. 38, no. 18, pp. 3831–3843, Jun. 1999.
[48] S. Sabour, N. Frosst, and G. E. Hinton, "Dynamic routing between capsules," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 3856–3866.
[49] K. Zhu, Y. Chen, P. Ghamisi, X. Jia, and J. A. Benediktsson, "Deep convolutional capsule network for hyperspectral image spectral and spectral–spatial classification," *Remote Sens.*, vol. 11, no. 3, p. 223, Jan. 2019.
[50] X. Wang, K. Tan, P. Du, B. Han, and J. Ding, "A capsule-vectored neural network for hyperspectral image classification," *Knowl.-Based Syst.*, vol. 268, May 2023, Art. no. 110482.
[51] H. Zhai and J. Zhao, "Two-stream spectral–spatial convolutional capsule network for hyperspectral image classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 127, Mar. 2024, Art. no. 103614.
[52] L. Sun, X. Song, H. Guo, G. Zhao, and J. Wang, "Patch-wise semantic segmentation for hyperspectral images via a cubic capsule network with EMAP features," *Remote Sens.*, vol. 13, no. 17, p. 3497, Sep. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.

[53] F. Deng, S. Pu, X. Chen, Y. Shi, T. Yuan, and S. Pu, "Hyperspectral image classification with capsule network using limited training samples," *Sensors*, vol. 18, no. 9, p. 3153, Sep. 2018.
[54] D. Kumar and D. Kumar, "A spectral–spatial 3D-convolutional capsule network for hyperspectral image classification with limited training samples," *Int. J. Inf. Technol.*, vol. 15, no. 1, pp. 379–391, Jan. 2023.
[55] L. Wei, H. Ran, Y. Yin, and H. Yang, "Multi-scale depthwise separable capsule network for hyperspectral image classification," *PLoS ONE*, vol. 19, no. 8, Aug. 2024, Art. no. e0308789.
[56] R. Giri, D. R. Pant, J. Heikkonen, and R. Kanth, "Hybrid capsule network for hyperspectral image unmixing and classification," in *Proc. Int. Conf. Adv. Comput. Res.* Cham, Switzerland: Springer, Jan. 2023, pp. 155–168.
[57] W. Wang, Y. Xu, Z. Xu, C. Kong, N. Xuemei, and J. Huang, "Multiscale spectral–spatial capsule neural network for hyperspectral image classification," in *Proc. Int. Conf. Efficiency Perform. Eng. Netw.* Cham, Switzerland: Springer, Jan. 2024, pp. 185–194.
[58] Q. Xu, Y. Tang, and Y. She, "Unsupervised multi-branch capsule for hyperspectral and LiDAR classification," *Proc. SPIE*, vol. 12617, pp. 70–79, Jul. 2023.
[59] N. Wang, A. Yang, Z. Cui, Y. Ding, Y. Xue, and Y. Su, "Capsule attention network for hyperspectral image classification," *Remote Sens.*, vol. 16, no. 21, p. 4001, Oct. 2024.
[60] M. Shi, R. Wang, and J. Ren, "Hierarchical capsule network for hyperspectral image classification," *Neural Comput. Appl.*, vol. 35, no. 25, pp. 18417–18443, Sep. 2023.



Qi Li received the B.S. degree from Xi'an University of Technology, Xi'an, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an.

His research interests include image processing, machine learning, and remote sensing.



Junwen Wang received the B.S. degree from Northwestern Polytechnical University, Xi'an, China, in 2024, where he is currently pursuing the M.S. degree with the School of Electronics and Information.

His research interests include hyperspectral image processing, machine learning, and deep learning.



Xingyuan Zu received the B.S. degree from South China Agricultural University, Guangzhou, China, in 2023. He is currently pursuing the M.S. degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His research interests include deep learning, computer vision, and hyperspectral image processing.



Dan Chen received the B.S. degree from the Northwestern Polytechnical University, Xi'an, China, in 2005, and the Ph.D. degree from Xi'an University of Technology, Xi'an, in 2012.

She is currently a Professor with the School of Automation and Information Engineering, Xi'an University of Technology. From 2015 to 2016, she was a Visiting Scholar with Ohio State University, Columbus, OH, USA. Her research interests include optical communication and robot intelligent information processing.



Ke Zhang received the B.S. degree from Northwestern Polytechnical University, Xi'an, China, in 2006, and the Ph.D. degree from Beijing Institute of Technology, Beijing, China, in 2023.

He is currently the Director of the Science and Technology on Electromechanical Dynamic Control Laboratory, Xi'an. His research interests include signal processing and image processing.



Jinghua Li was born in Shanxi, China, in 1964. She received the B.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, China, in 1987 and 2008, respectively.

She is currently a Full Professor with the School of Electronics and Information, Northwestern Polytechnical University. Her research interests include image processing, digital signal processing, and pattern recognition, with applications in the field of remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:30:08 UTC from IEEE Xplore. Restrictions apply.